



Autumn Examinations 2009/ 2010

Exam Code(s) 1MF1, 1SD1, 2CS1
Exam(s) Higher Diploma in Applied Science (Software Design & Development), M.Sc in Software Design & Development, BSc (Computing Studies)

Module Code(s) CT853, CT232
Module(s) Algorithmics & Logical Methods
Methodology

Paper No. 1

External Examiner(s) Prof. Michael O'Boyle
Internal Examiner(s) Prof. Gerard Lyons.
Dr Jim Duggan
Dr Finlay Smith

Instructions:

Answer any **3** questions

All questions carry equal marks.

Duration 2 hours
No. of Pages 4
Department(s) IT

You do NOT need to give any code in your answers. You should describe or show (as appropriate) the effect of the code.

Q.1.

(i) *5 marks*

What operations would be required in order to implement a stack? What techniques could be used to implement a stack? Highlight the advantages & disadvantages of each technique.

(ii) *15 marks*

Use a stack to check the validity of the following sequences of brackets (show each stage of the process):

a) [{ () () } { } { { { } } }]

b) { { { { [{ (()) }] } } } }

Q.2.

(i) *8 marks*

What operations would be required to implement the data type queue? Describe the problems associated with implementing queues as an array, explain how these problems can be overcome. Illustrate your answer with examples.

(ii) *8 marks*

Show how the following operations are performed on an empty queue for both array implementation and linked implementation:

add 'A'

add 'B'

remove element

add 'C'

add 'D'

add 'E'

remove element

destroy queue

(iii) *4 marks*

Give 4 examples of situations where a computerised queue could be used.

Q.3.

(i) *6 marks*

Explain what is meant by the following terms:

- a) Binary tree
- b) Full tree
- c) Complete tree
- d) pre-order tree parsing

(ii) *8 marks*

Show the steps involved in creating a binary search tree with the following data (in the order given):

G C A B F D I K H L M N

Show how the tree would be searched for the value E.

(iii) *6 marks*

repeat (ii) but this time insert the elements in reverse order.

Q.4.

(i) *5 marks*

Explain what is meant by an Abstract Data Type. Highlight the advantages of using them.

(ii) *5 marks*

Describe three different possible implementations of the Abstract Data Type **List**. Give a list of circumstances where each of them would be the preferred implementation. Explain your answer.

(iii) *10 marks*

Draw a linked list that contains the elements 'B', 'C', 'A' and 'D' (in that order).

with your linked list show the steps involved in the following operations (for each step assume that the previous steps have been executed):

Move to the first element in the list

Move to the next element in the list

Delete the current element

Insert the value 'E'

At each step you should show all of the changes that are required and also show the changes to any variables.